ModelIt!

Teacher's Guide

# Potential to Kinetic: The Energy Transformation Chain

How Height, Mass, and Friction Determine Where Energy Goes

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 08L2 |
| **NGSS Standard** | MS-PS3-1, MS-PS3-5 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

# Table of Contents

ModelIt!


1 NGSS Standards Unpacking & CAST Alignment ..... 3

2 Scoring Rubric ..... 4

3 Background Content Knowledge ..... 5

4 The LEVER Framework ..... 6

5 Slide-by-Slide Facilitation Guide ..... 7

6 Answer Key & Expected Responses ..... 9

7 Collaborative Learning Protocols ..... 10

8 Differentiation & Extensions ..... 11

9 Common Misconceptions ..... 12

# 1 NGSS Standards & CAST Alignment

ModelIt!

**Performance Expectation: MS-PS3-1, MS-PS3-5**

Construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass of an object and to the speed of an object. Construct, use, and present arguments to support the claim that when the kinetic energy of an object changes, energy is transferred to or from the object.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Developing and Using Models** — Students build a model showing energy transformations between potential, kinetic, and thermal forms while demonstrating conservation. |
| **Disciplinary Core Idea** | **PS3.B Conservation of Energy** — Energy is transferred between objects or converted between forms, but the total energy in a closed system is conserved. |
| **Crosscutting Concept** | **Energy and Matter** — Students track energy through transformations, demonstrating that energy is conserved in quantity even as it changes form. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A 2 kg ball is released from a height of 10 meters on a frictionless ramp. At the bottom, its kinetic energy is 196 J. If the same ball is released on a ramp with friction, at the bottom its kinetic energy is 150 J. What happened to the other 46 J?

**CONSTRUCTED RESPONSE:**

Using evidence from your model, explain why a heavier object released from the same height as a lighter object reaches the bottom with more kinetic energy. Include the role of potential energy and conservation of energy in your explanation.

# 2 Scoring Rubric

ModelIt!

### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

ModelIt!

**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Energy transformation is one of the most fundamental concepts in physics. Every motion, every machine, every living process involves energy changing from one form to another. Understanding these transformations and the law of conservation of energy is essential for engineering, technology, and understanding the natural world.

## Potential and Kinetic Energy

Gravitational potential energy depends on two factors: mass and height (PE = mgh, where m is mass, g is gravitational acceleration at 9.8 $m/s^2$, and h is height). Kinetic energy depends on mass and speed (KE = $\frac{1}{2}mv^2$). As an object falls, height decreases (PE drops) while speed increases (KE rises). The conversion is continuous and predictable.

## The Conservation Law

The law of conservation of energy states that energy cannot be created or destroyed — only transformed. In a closed system, total energy remains constant. This law was established through the work of James Joule, Hermann von Helmholtz, and Julius Robert von Mayer in the 1840s. It is one of the most fundamental laws of physics and has never been violated.

## Friction and Thermal Energy

Friction is not an energy-destroyer but an energy-transformer. When surfaces rub together, kinetic energy converts to thermal energy (heat) at the molecular level. The 'lost' energy is actually in the form of faster-vibrating molecules at the contact surfaces. This is why rubbing your hands together makes them warm. In engineering, friction is managed, not eliminated.

## Real-World Energy Chains

A car engine converts chemical energy (gasoline) to thermal energy (combustion) to kinetic energy (motion) — with inevitable losses to heat and sound at each step. A hydroelectric dam converts gravitational PE (water at height) to KE (falling water) to electrical energy (generator). Every energy system is a chain of transformations governed by conservation.

# 4 The LEVER Framework

ModelIt!

### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L — Locate the System**
Students identify height, mass, potential energy, kinetic energy, and thermal energy as system components.

**E — Establish Relationships**
Students determine that height and mass both positively affect potential energy, PE converts to KE as objects fall, and friction converts KE to thermal energy.

**V — Visualize & Model**
Students build the energy transformation model showing PE → KE → Thermal while total energy remains constant.

**E — Evaluate Outcomes**
Students compare ideal (frictionless) vs. friction scenarios and verify that PE + KE + Thermal always equals the starting energy.

**R — Revise & Extend**
Students add air resistance or elastic energy components to model more complex energy chains.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

ModelIt!

## Slide 1: Cover

2 min

**Say:** *"Right now, this roller coaster has more energy than at any other point on the ride. How?"*

**Watch For**

Show a roller coaster photo. Ask: Where is the energy stored?

## Slide 2: Learning Objectives

3 min

**Say:** *"Today we track energy as it transforms from stored to moving to heat."*

**Watch For**

Pre-teach 'potential' (stored, waiting) vs. 'kinetic' (moving, active).

## Slide 3: Big Question

3 min

**Say:** *"If a ball slows down at the bottom of a ramp, where did the energy go? It can't just vanish."*

**Watch For**

Quick demo: roll a ball on a smooth surface vs. carpet. Where did the speed go?

## Slide 4: LEVER Framework

2 min

**Say:** *"We'll build a model to track every joule of energy from start to finish."*

**Watch For**

Preview that total energy will NEVER change — only its form.

## Slide 5: Activity 1: Components

8 min

**Say:** *"Height and mass are our inputs. PE, KE, and thermal energy are the outputs."*

**Watch For**

Sort components. Discuss why height and mass are external (we set them).

## Slide 6: Activity 2: Connections

8 min

**Say:** *"Energy flows: PE becomes KE, KE becomes thermal. But the total never changes."*

**Watch For**

Map relationships. Draw the transformation chain. Highlight the conservation rule.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Watch the bars trade places: PE shrinks, KE grows. Add friction — thermal appears."*

**Watch For**

Students predict bar graph changes. Run ideal vs. friction scenarios. Verify total stays constant.

## Slide 8: Discoveries

5 min

**Say:** *"Energy is the ultimate shape-shifter. It changes form but never amount."*

**Watch For**

Sum PE + KE + Thermal at multiple time points. They always equal the starting PE.

## Slide 9: STEM Challenge

12 min

**Say:** *"Engineers fight friction every day. Design a ramp that delivers maximum KE to the bottom."*

**Watch For**

Teams design, test with model, and present. Compare efficiency ratings.

# 6 Answer Key & Expected Responses

ModelIt!

## Component Sorting

- **EXTERNAL:** Height, Mass
- **INTERNAL:** Potential Energy, Kinetic Energy, Thermal Energy (Friction)

## Relationships

- **Height to Potential Energy:** POSITIVE (+) — Greater height means more gravitational potential energy stored in the object (PE = mgh).
- **Mass to Potential Energy:** POSITIVE (+) — Greater mass means more gravitational potential energy at the same height (PE = mgh).
- **Potential Energy to Kinetic Energy:** POSITIVE (+) — As the object descends, stored potential energy transforms into kinetic energy of motion.
- **Kinetic Energy to Thermal Energy:** POSITIVE (+) — Friction between the moving object and surfaces converts some kinetic energy into heat.
- **Thermal Energy to Kinetic Energy:** NEGATIVE (-) — Energy lost to friction as heat reduces the kinetic energy available for motion — the object slows down.

## Key Discoveries

- As an object falls, its potential energy decreases while kinetic energy increases — energy transforms, not disappears
- Heavier objects store more potential energy at the same height and therefore have more kinetic energy at the bottom
- Friction converts kinetic energy into thermal energy (heat), slowing the object down
- Total energy (PE + KE + Thermal) remains CONSTANT — conservation of energy is always obeyed
- In a frictionless world, PE at the top would exactly equal KE at the bottom

### Multiple Choice Post-Assessment Answer Key

**1. Your model shows that a 5 kg ball released from 10 meters has 490 J of potential energy at the top and 490 J of kinetic energy at the bottom on a frictionless ramp. On a ramp with friction, the ball reaches the bottom with only 380 J of kinetic energy. Where are the other 110 J?**

**Correct: B) The 110 J were converted to thermal energy (heat) in the ramp surface and ball through friction**

*Correct! Energy is never destroyed. Friction converts kinetic energy into thermal energy (heat) at the contact surfaces. The total energy is still 490 J: 380 J kinetic + 110 J thermal = 490 J. Conservation of energy is maintained.*

**2. In your model, a 2 kg object and a 8 kg object are released from the same height. The model shows the 8 kg object has four times more kinetic energy at the bottom. A student concludes the heavier object must fall four times faster. Is this correct?**

**Correct: B) No. The heavier object has more kinetic energy because KE depends on mass AND speed. Both objects fall at the same speed (without air resistance), but the heavier one has more energy due to its greater mass.**

*Correct! KE = 1/2 mv squared. The heavier object has four times the mass but the same speed (gravity accelerates all objects equally). Four times the mass at the same speed gives four times the kinetic energy. Mass affects energy, not fall rate.*

**3. The model shows that at the midpoint of a frictionless ramp, an object has both potential and kinetic energy. Which statement is true about the total energy at this point?**

**Correct: B) Total energy equals the PE at the top, split between PE (remaining height) and KE (gained speed)**

*Correct! At the midpoint, the object has descended half the height (losing half its PE) and gained speed (gaining KE equal to the lost PE). Total energy is unchanged: PE at midpoint + KE at midpoint = original PE at the top. This holds at every point.*

**4. Based on your model, why can a roller coaster's second hill never be taller than its first hill?**

**Correct: B) Because friction and air resistance convert some kinetic energy to thermal energy along the way, leaving insufficient energy to reach the original height**

*Correct! The first hill sets the total energy budget. Friction and air resistance convert some KE to thermal energy during the ride. Each subsequent hill has less available energy than the previous one. There is never enough energy remaining to reach the original height.*

**5. A student claims that a ball rolling on carpet 'loses energy' compared to rolling on a smooth floor. Using model evidence, how should this claim be corrected?**

**Correct: B) The ball does not lose energy. Carpet has higher friction, which converts more kinetic energy to thermal energy. The total energy (KE + thermal) remains the same as on the smooth floor.**

*Correct! The model proves that total energy (PE + KE + Thermal) is always conserved. Carpet has higher friction, converting more KE to thermal energy, so the ball slows down faster. But the energy is not lost; it becomes heat in the carpet fibers. Energy transforms, never disappears.*

# 7 Collaborative Learning Protocols

ModelIt!

### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

ModelIt!

### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

ModelIt!

### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Energy is used up or consumed" | Energy is NEVER used up — it transforms. When a ball 'loses' speed to friction, that kinetic energy becomes thermal energy in the surface. The total energy hasn't changed; it's just in a less useful form. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Heavier objects fall faster" | In a vacuum (no air resistance), all objects fall at the same rate regardless of mass. Heavier objects have more PE and more KE, but they also need more force to accelerate. The effects cancel out. Mass affects energy, not fall rate. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Friction destroys energy" | Friction transforms kinetic energy into thermal energy — it doesn't destroy it. The heat generated is real energy that warms the surfaces. In the model, adding friction reduces KE but adds exactly that much thermal energy. The total stays constant. |